

Requirements:

Features Multi-language Support
French, Japanese, German, Spanish, Korean (initial set)
CEFR-aligned levels (A1, A2, B1)
Grade-wise Curriculum Tailored lessons for Classes 6 to 10
Mapped with CBSE/ICSE/International boards
Gamified Learning
Quizzes and vocabulary games
Points, leaderboards, badges, and rewards
Daily streaks and goal tracking
Live Interactive Classes
Scheduled live classes with language experts
Small batch size for student-teacher interaction
Self-paced Recorded Lessons
Bite-sized video lectures
Thematic lessons based on real-life scenarios
AI-powered Language Lab
Speech-to-text for pronunciation accuracy
Text-to-speech practice for fluency
Real-time feedback on speaking
Interactive Assessments
End-of-module quizzes
Reading, listening, speaking, and writing tasks
Progress Tracking Dashboard Weekly & monthly reports
Performance analytics and milestones
Certification CEFR-aligned certificates for each level completed
Shareable on LinkedIn and resumes
Cultural Immersion Modules Country-specific content (festivals, food, etiquette)
Videos, games, and storytelling formats
Custom LMS (Learning Management System) Lesson plans, calendar, and assignment tracker Integrated with chat & doubt forums
Parent & Teacher Portals
Progress updates for parents
School integration tools for teachers
Mobile App Access Android & iOS app for on-the-go learning
Offline content access Multilingual UI Support
Platform interface available in English and regional languages for ease
Community Features
Student discussion forums

Language buddy programs

1. Architecture Overview

Frontend:

- **Web (Responsive):** React.js + Tailwind CSS
- **Mobile:** React Native (for cross-platform support – Android & iOS)

Backend:

- **Node.js with Express.js**
- **Authentication:** Firebase Auth or Auth0
- **Database:**
 - **MongoDB** for flexibility (especially for user progress, gamification)

AI Components:

- **Speech-to-Text:** Google Cloud Speech API or AssemblyAI
- **Text-to-Speech:** Google TTS or Amazon Polly
- **Pronunciation Feedback:** Whisper + custom ML (optional)

Chatbot + Live Chat:

- **Dialogflow** or **Botpress** for chatbot
- **Live Chat:** Socket.IO for real-time chat; scheduling logic for 10-min chat + Stripe for premium upgrade

Gamification Engine:

- Custom module with:
 - XP points, streaks
 - Leaderboard logic stored in Redis
 - Badge assignments via backend

Admin Dashboard (LMS):

- **React.js Admin Panel**
- Role-based access (Admin, Teacher, Parent, Student)
- Charts: Recharts or Chart.js

CI/CD & Deployment:

- Vercel (for frontend) + Render for backend
- GitHub Actions for CI/CD
- Firebase Hosting (optional for web app)

Mobile App:

- React Native + Firebase or local SQLite for offline mode
- Push notifications via Firebase Cloud Messaging (FCM)

2. Requirements from NTS

Domain & Hosting	- Subdomain for dev/testing
Content	- Language curriculum (mapped to CEFR & grades)
Design Input	- Logos, color palette, branding
Live Class Infra	- Zoom API credentials or custom video platform access
Premium Features	- Stripe/Razorpay account access for integration
Teacher Chat Logic	- Availability schedule APIs
Voice Model Budget	- Google Cloud / Whisper / AssemblyAI billing support
Multilingual Content	- Help translating platform UI & courses
Test Users	- Dummy students/teachers for testing live chat, gamification